

Trailing factors: normalized z (signal) over absolute avg (context), by forward book-move segment

SINCE COVID (2020-04-01 to 2026-06-18, n=1562) | z = rolling z-score (value - recent mean)/recent std, L=63d

Each cell: TOP = mean z in-segment, GREEN=above recent norm / RED=below, deeper=stronger signal. BOTTOM (grey) = absolute avg (context).

D = UP minus DN (z on top colored; absolute gap grey below). Returns/vol abs in %; scores raw.

| TRAILING FACTOR     | FWD 1D      |          |        | FWD 5D      |          |        | FWD 20D     |          |        |
|---------------------|-------------|----------|--------|-------------|----------|--------|-------------|----------|--------|
|                     | thr +/-2.0% |          |        | thr +/-4.5% |          |        | thr +/-7.5% |          |        |
|                     | UP n=230    | DN n=183 | D(U-D) | UP n=269    | DN n=162 | D(U-D) | UP n=422    | DN n=161 | D(U-D) |
|                     |             |          |        |             |          |        |             |          |        |
| BOOK                |             |          |        |             |          |        |             |          |        |
| book_1d             | -0.11       | -0.09    | -0.02  | -0.10       | +0.03    | -0.13  | +0.09       | +0.06    | +0.03  |
|                     | -0.1%       | +0.1%    | -0.2   | -0.1%       | +0.2%    | -0.2   | +0.3%       | +0.2%    | +0.1   |
| book_5d             | -0.22       | -0.12    | -0.10  | -0.19       | -0.12    | -0.06  | +0.13       | +0.00    | +0.13  |
|                     | +0.0%       | +0.4%    | -0.4   | +0.0%       | +0.2%    | -0.2   | +1.1%       | +0.6%    | +0.5   |
| book_20d            | +0.01       | -0.08    | +0.09  | +0.13       | -0.21    | +0.34  | +0.25       | -0.29    | +0.54  |
|                     | +3.6%       | +2.7%    | +0.9   | +3.2%       | +1.4%    | +1.8   | +4.0%       | +0.5%    | +3.5   |
| LONG/SHORT SPREAD   |             |          |        |             |          |        |             |          |        |
| spread_1d           | -0.14       | -0.05    | -0.09  | +0.03       | +0.08    | -0.05  | +0.07       | +0.12    | -0.05  |
|                     | -0.3%       | -0.1%    | -0.2   | +0.0%       | +0.1%    | -0.1   | +0.1%       | +0.2%    | -0.1   |
| spread_5d           | +0.00       | +0.06    | -0.05  | +0.06       | -0.01    | +0.08  | +0.14       | +0.01    | +0.13  |
|                     | -0.0%       | +0.2%    | -0.2   | -0.0%       | -0.3%    | +0.3   | +0.3%       | -0.4%    | +0.7   |
| spread_20d          | +0.01       | +0.05    | -0.04  | +0.15       | +0.01    | +0.14  | +0.07       | +0.08    | -0.01  |
|                     | +0.2%       | -0.0%    | +0.2   | +0.6%       | -0.6%    | +1.2   | -0.2%       | -0.6%    | +0.4   |
| BASKETS             |             |          |        |             |          |        |             |          |        |
| top10_1d            | -0.15       | -0.12    | -0.04  | -0.04       | +0.03    | -0.07  | +0.09       | +0.11    | -0.01  |
|                     | -0.2%       | -0.1%    | -0.1   | -0.0%       | +0.2%    | -0.1   | +0.3%       | +0.4%    | -0.0   |
| top10_5d            | -0.16       | -0.07    | -0.09  | -0.09       | -0.06    | -0.03  | +0.14       | +0.12    | +0.02  |
|                     | +0.1%       | +0.5%    | -0.3   | +0.2%       | +0.5%    | -0.3   | +1.3%       | +1.1%    | +0.2   |
| top10_20d           | -0.04       | -0.05    | +0.01  | +0.10       | -0.10    | +0.20  | +0.26       | +0.05    | +0.21  |
|                     | +3.4%       | +2.7%    | +0.7   | +3.5%       | +2.0%    | +1.5   | +4.4%       | +1.8%    | +2.6   |
| bottom10_1d         | -0.06       | -0.06    | -0.00  | -0.08       | -0.09    | +0.01  | +0.01       | -0.06    | +0.07  |
|                     | +0.1%       | +0.0%    | +0.1   | +0.0%       | +0.0%    | +0.0   | +0.2%       | +0.2%    | +0.1   |
| bottom10_5d         | -0.22       | -0.21    | -0.01  | -0.14       | -0.17    | +0.03  | -0.04       | +0.03    | -0.07  |
|                     | +0.2%       | +0.3%    | -0.1   | +0.2%       | +0.8%    | -0.5   | +1.0%       | +1.5%    | -0.4   |
| bottom10_20d        | -0.13       | -0.19    | +0.06  | -0.10       | -0.24    | +0.14  | +0.03       | -0.15    | +0.18  |
|                     | +3.2%       | +2.7%    | +0.5   | +2.9%       | +2.6%    | +0.3   | +4.6%       | +2.4%    | +2.3   |
| universe_1d         | -0.16       | -0.09    | -0.07  | -0.05       | -0.00    | -0.05  | +0.05       | -0.00    | +0.05  |
|                     | -0.1%       | +0.0%    | -0.1   | +0.0%       | +0.1%    | -0.1   | +0.2%       | +0.2%    | +0.0   |
| universe_5d         | -0.23       | -0.20    | -0.03  | -0.14       | -0.16    | +0.02  | +0.03       | +0.05    | -0.03  |
|                     | +0.1%       | +0.3%    | -0.2   | +0.1%       | +0.5%    | -0.4   | +0.9%       | +1.2%    | -0.3   |
| universe_20d        | -0.14       | -0.23    | +0.09  | -0.01       | -0.27    | +0.26  | +0.12       | -0.13    | +0.24  |
|                     | +2.5%       | +2.1%    | +0.4   | +2.3%       | +1.8%    | +0.5   | +3.5%       | +2.4%    | +1.2   |
| ETFs                |             |          |        |             |          |        |             |          |        |
| spy_1d              | -0.15       | +0.01    | -0.16  | -0.07       | +0.04    | -0.11  | +0.02       | +0.00    | +0.02  |
|                     | -0.1%       | +0.1%    | -0.2   | +0.0%       | +0.1%    | -0.1   | +0.1%       | +0.1%    | +0.0   |
| spy_5d              | -0.22       | -0.18    | -0.04  | -0.17       | -0.10    | -0.07  | -0.01       | +0.06    | -0.07  |
|                     | +0.1%       | +0.2%    | -0.1   | +0.0%       | +0.3%    | -0.3   | +0.4%       | +0.5%    | -0.1   |
| spy_20d             | -0.25       | -0.33    | +0.09  | -0.15       | -0.29    | +0.14  | -0.02       | -0.04    | +0.03  |
|                     | +1.1%       | +1.0%    | +0.1   | +0.9%       | +0.9%    | -0.0   | +1.7%       | +1.3%    | +0.5   |
| qqq_1d              | -0.16       | -0.02    | -0.15  | -0.05       | +0.03    | -0.09  | +0.02       | +0.00    | +0.02  |
|                     | -0.1%       | +0.1%    | -0.2   | +0.0%       | +0.1%    | -0.1   | +0.1%       | +0.1%    | +0.0   |
| qqq_5d              | -0.22       | -0.14    | -0.08  | -0.15       | -0.12    | -0.03  | -0.02       | +0.02    | -0.04  |
|                     | +0.0%       | +0.2%    | -0.2   | -0.0%       | +0.3%    | -0.3   | +0.5%       | +0.6%    | -0.1   |
| qqq_20d             | -0.23       | -0.27    | +0.04  | -0.13       | -0.26    | +0.13  | -0.03       | -0.12    | +0.09  |
|                     | +1.4%       | +1.3%    | +0.1   | +1.2%       | +1.1%    | +0.1   | +2.2%       | +1.5%    | +0.7   |
| smh_1d              | -0.16       | -0.02    | -0.14  | +0.01       | +0.06    | -0.05  | +0.05       | +0.02    | +0.03  |
|                     | -0.1%       | +0.2%    | -0.3   | +0.2%       | +0.3%    | -0.1   | +0.3%       | +0.2%    | +0.1   |
| smh_5d              | -0.17       | -0.08    | -0.10  | -0.10       | -0.10    | -0.00  | +0.01       | +0.05    | -0.03  |
|                     | +0.3%       | +0.7%    | -0.5   | +0.3%       | +0.5%    | -0.2   | +0.9%       | +0.9%    | +0.0   |
| smh_20d             | -0.10       | -0.18    | +0.08  | -0.09       | -0.24    | +0.15  | -0.04       | -0.20    | +0.17  |
|                     | +3.5%       | +3.2%    | +0.3   | +2.6%       | +1.9%    | +0.7   | +3.3%       | +1.9%    | +1.4   |
| psq_1d              | +0.17       | +0.01    | +0.15  | +0.05       | -0.04    | +0.09  | -0.02       | +0.00    | -0.02  |
|                     | +0.1%       | -0.1%    | +0.2   | +0.0%       | -0.1%    | +0.1   | -0.1%       | -0.1%    | -0.0   |
| psq_5d              | +0.22       | +0.14    | +0.09  | +0.16       | +0.12    | +0.04  | +0.02       | -0.02    | +0.04  |
|                     | +0.1%       | -0.2%    | +0.2   | +0.1%       | -0.2%    | +0.3   | -0.4%       | -0.5%    | +0.1   |
| psq_20d             | +0.24       | +0.28    | -0.03  | +0.15       | +0.26    | -0.11  | +0.04       | +0.14    | -0.10  |
|                     | -1.2%       | -0.9%    | -0.2   | -1.0%       | -0.8%    | -0.1   | -1.9%       | -1.1%    | -0.8   |
| soxs_1d             | +0.16       | +0.02    | +0.14  | -0.02       | -0.07    | +0.05  | -0.04       | -0.02    | -0.02  |
|                     | +0.5%       | -0.5%    | +1.0   | -0.5%       | -0.0%    | +0.3   | -0.7%       | -0.6%    | -0.1   |
| soxs_5d             | +0.17       | +0.04    | +0.13  | +0.12       | +0.07    | +0.05  | +0.02       | -0.07    | +0.08  |
|                     | -0.7%       | -2.7%    | +2.1   | -0.7%       | -1.9%    | +1.2   | -2.4%       | -2.9%    | +0.5   |
| soxs_20d            | +0.10       | +0.15    | -0.05  | +0.16       | +0.18    | -0.02  | +0.10       | +0.13    | -0.03  |
|                     | -10.9%      | -9.8%    | -1.1   | -8.2%       | -7.2%    | -1.0   | -9.9%       | -7.1%    | -2.9   |
| AVG SCORE           |             |          |        |             |          |        |             |          |        |
| sc.top10_1d         | +0.03       | +0.04    | -0.01  | +0.02       | -0.12    | +0.14  | +0.11       | -0.18    | +0.30  |
|                     | 0.841       | 0.841    | -0.000 | 0.839       | 0.836    | +0.003 | 0.842       | 0.834    | +0.007 |
| sc.top10_5d         | +0.01       | +0.02    | -0.01  | +0.03       | -0.15    | +0.18  | +0.11       | -0.35    | +0.46  |
|                     | 0.840       | 0.840    | +0.001 | 0.839       | 0.836    | +0.003 | 0.840       | 0.833    | +0.008 |
| sc.top10_20d        | +0.07       | -0.03    | +0.10  | +0.10       | -0.30    | +0.40  | +0.10       | -0.33    | +0.42  |
|                     | 0.841       | 0.840    | +0.001 | 0.840       | 0.838    | +0.002 | 0.840       | 0.838    | +0.001 |
| sc.bottom10_1d      | -0.03       | -0.00    | -0.03  | +0.10       | +0.22    | -0.12  | +0.07       | +0.14    | -0.07  |
|                     | 0.206       | 0.205    | +0.001 | 0.210       | 0.212    | -0.002 | 0.208       | 0.209    | -0.001 |
| sc.bottom10_5d      | -0.09       | -0.09    | +0.08  | +0.11       | +0.15    | -0.04  | +0.13       | +0.03    | +0.10  |
|                     | 0.206       | 0.204    | +0.002 | 0.209       | 0.209    | +0.000 | 0.208       | 0.206    | +0.002 |
| sc.bottom10_20d     | +0.09       | -0.02    | +0.11  | +0.15       | -0.00    | +0.15  | +0.22       | +0.00    | +0.22  |
|                     | 0.207       | 0.206    | +0.001 | 0.208       | 0.207    | +0.001 | 0.207       | 0.206    | +0.002 |
| sc.universe_1d      | -0.41       | -0.38    | -0.04  | -0.28       | -0.39    | +0.11  | -0.36       | -0.32    | -0.04  |
|                     | 0.506       | 0.506    | +0.000 | 0.506       | 0.506    | +0.000 | 0.506       | 0.506    | +0.000 |
| sc.universe_5d      | -0.40       | -0.50    | +0.10  | -0.39       | -0.62    | +0.23  | -0.46       | -0.33    | -0.14  |
|                     | 0.506       | 0.506    | +0.000 | 0.506       | 0.506    | +0.000 | 0.506       | 0.506    | +0.000 |
| sc.universe_20d     | -0.57       | -0.74    | +0.17  | -0.58       | -0.82    | +0.24  | -0.61       | -0.44    | -0.17  |
|                     | 0.506       | 0.506    | +0.000 | 0.506       | 0.506    | +0.000 | 0.506       | 0.506    | +0.000 |
| sc.held_1d          | +0.12       | +0.05    | +0.06  | +0.18       | -0.04    | +0.22  | +0.14       | -0.11    | +0.25  |
|                     | 0.611       | 0.608    | +0.002 | 0.612       | 0.600    | +0.012 | 0.601       | 0.594    | +0.007 |
| sc.held_5d          | +0.14       | +0.09    | +0.05  | +0.24       | +0.03    | +0.21  | +0.14       | -0.17    | +0.31  |
|                     | 0.610       | 0.609    | +0.001 | 0.612       | 0.603    | +0.009 | 0.599       | 0.594    | +0.005 |
| sc.held_20d         | +0.28       | +0.08    | +0.20  | +0.32       | -0.01    | +0.33  | +0.18       | -0.02    | +0.20  |
|                     | 0.611       | 0.607    | +0.004 | 0.607       | 0.601    | +0.006 | 0.595       | 0.602    | -0.007 |
| REALIZED VOL (ann.) |             |          |        |             |          |        |             |          |        |
| spy_vol_1d          | +0.07       | +0.07    | -0.00  | +0.06       | +0.18    | -0.12  | +0.03       | -0.03    | +0.06  |
|                     | +12.9%      | +13.0%   | -0.1   | +14.2%      | +13.5%   | +0.7   | +14.1%      | +11.9%   | +2.2   |
| spy_vol_5d          | +0.03       | +0.21    | -0.18  | +0.07       | +0.29    | -0.21  | +0.00       | -0.14    | +0.14  |
|                     | +15.6%      | +16.2%   | -0.6   | +17.1%      | +15.6%   | +1.5   | +16.8%      | +14.1%   | +2.8   |
| spy_vol_20d         | +0.07       | +0.08    | -0.02  | +0.13       | +0.08    | +0.05  | +0.11       | -0.12    | +0.23  |
|                     | +17.5%      | +16.0%   | +1.5   | +19.0%      | +15.1%   | +3.9   | +18.1%      | +15.2%   | +2.8   |
| qqq_vol_1d          | +0.09       | +0.05    | +0.05  | +0.09       | +0.13    | -0.03  | +0.01       | +0.03    | -0.03  |
|                     | +17.6%      | +17.1%   | +0.5   | +18.7%      | +17.9%   | +0.7   | +17.3%      | +17.2%   | +0.1   |
| qqq_vol_5d          | +0.08       | +0.23    | -0.15  | +0.13       | +0.26    | -0.13  | -0.01       | -0.01    | -0.00  |
|                     | +21.0%      | +21.9%   | -0.9   | +22.2%      | +21.1%   | +1.1   | +20.9%      | +20.3%   | +0.6   |
| qqq_vol_20d         | +0.17       | +0.20    | -0.03  | +0.20       | +0.19    | +0.01  | +0.16       | +0.12    | +0.03  |
|                     | +22.8%      | +21.6%   | +1.2   | +23.9%      | +21.1%   | +2.8   | +22.3%      | +21.6%   | +0.8   |

z color = signal (green above norm / red below, darker=stronger). Grey = raw level for context only. In-segment z is range-compressed by tail conditioning.

Biggest UP-DN gaps, sorted by |z-gap| within each horizon (z=signal, abs=context)

SINCE COVID (2020-04-01 to 2026-06-18, n=1562) | z = rolling z-score (value - recent mean)/recent std, L=63d

Sorted by absolute normalized gap. Dz colored (green/red, deeper=stronger); abs gap in grey.

FWD 1D (thr +/-2.0%)

| factor             | Dz (absGap   zUP/zDN)      |
|--------------------|----------------------------|
| 1.sc.held_20d      | +0.20 +0.004   +0.28/+0.08 |
| 2.spy_vol_5d       | -0.18 -0.6   +0.03/+0.21   |
| 3.sc.universe_20d  | +0.17 +0.000   -0.57/-0.74 |
| 4.spy_1d           | -0.16 -0.2   -0.15/+0.01   |
| 5.psq_1d           | +0.15 +0.2   +0.17/+0.01   |
| 6.qqq_vol_5d       | -0.15 -0.9   +0.08/+0.23   |
| 7.qqq_1d           | -0.15 -0.2   -0.16/-0.02   |
| 8.soxs_1d          | +0.14 +1.0   +0.16/+0.02   |
| 9.smh_1d           | -0.14 -0.3   -0.16/-0.02   |
| 10.soxs_5d         | +0.13 +2.1   +0.17/+0.04   |
| 11.sc.bottom10_20d | +0.11 +0.001   +0.09/-0.02 |
| 12.sc.universe_5d  | +0.10 +0.000   -0.40/-0.50 |
| 13.sc.top10_20d    | +0.10 +0.001   +0.07/-0.03 |
| 14.smh_5d          | -0.10 -0.5   -0.17/-0.08   |
| 15.book_5d         | -0.10 -0.4   -0.22/-0.12   |
| 16.spread_1d       | -0.09 -0.2   -0.14/-0.05   |
| 17.book_20d        | +0.09 +0.9   +0.01/-0.08   |
| 18.psq_5d          | +0.09 +0.2   +0.22/+0.14   |
| 19.universe_20d    | +0.09 +0.4   -0.14/-0.23   |
| 20.spy_20d         | +0.09 +0.1   -0.25/-0.33   |
| 21.top10_5d        | -0.09 -0.3   -0.16/-0.07   |
| 22.smh_20d         | +0.08 +0.3   -0.10/-0.18   |
| 23.qqq_5d          | -0.08 -0.2   -0.22/-0.14   |
| 24.sc.bottom10_5d  | +0.08 +0.002   -0.02/-0.09 |
| 25.universe_1d     | -0.07 -0.1   -0.16/-0.09   |
| 26.sc.held_1d      | +0.06 +0.002   +0.12/+0.05 |
| 27.bottom10_20d    | +0.06 +0.5   -0.13/-0.19   |
| 28.spread_5d       | -0.05 -0.2   +0.00/+0.06   |
| 29.soxs_20d        | -0.05 -1.1   +0.10/+0.15   |
| 30.sc.held_5d      | +0.05 +0.001   +0.14/+0.09 |
| 31.qqq_vol_1d      | +0.05 +0.5   +0.09/+0.05   |
| 32.spread_20d      | -0.04 +0.2   +0.01/+0.05   |
| 33.spy_5d          | -0.04 -0.1   -0.22/-0.18   |
| 34.top10_1d        | -0.04 -0.1   -0.15/-0.12   |
| 35.sc.universe_1d  | -0.04 +0.000   -0.41/-0.38 |
| 36.qqq_20d         | +0.04 +0.1   -0.23/-0.27   |
| 37.psq_20d         | -0.03 -0.2   +0.24/+0.28   |
| 38.qqq_vol_20d     | -0.03 +1.2   +0.17/+0.20   |
| 39.universe_5d     | -0.03 -0.2   -0.23/-0.20   |
| 40.sc.bottom10_1d  | -0.03 +0.001   -0.03/-0.00 |
| 41.book_1d         | -0.02 -0.2   -0.11/-0.09   |
| 42.spy_vol_20d     | -0.02 +1.5   +0.07/+0.08   |
| 43.bottom10_5d     | -0.01 -0.1   -0.22/-0.21   |
| 44.sc.top10_1d     | -0.01 -0.000   +0.03/+0.04 |
| 45.sc.top10_5d     | -0.01 +0.001   +0.01/+0.02 |
| 46.top10_20d       | +0.01 +0.7   -0.04/-0.05   |
| 47.spy_vol_1d      | -0.00 -0.1   +0.07/+0.07   |
| 48.bottom10_1d     | -0.00 +0.1   -0.06/-0.06   |

FWD 5D (thr +/-4.5%)

| factor             | Dz (absGap   zUP/zDN)      |
|--------------------|----------------------------|
| 1.sc.top10_20d     | +0.40 +0.002   +0.10/-0.30 |
| 2.book_20d         | +0.34 +1.8   +0.13/-0.21   |
| 3.sc.held_20d      | +0.33 +0.006   +0.32/-0.01 |
| 4.universe_20d     | +0.26 +0.5   -0.01/-0.27   |
| 5.sc.universe_20d  | +0.24 +0.000   -0.58/-0.82 |
| 6.sc.universe_5d   | +0.23 +0.000   -0.39/-0.62 |
| 7.sc.held_1d       | +0.22 +0.012   +0.18/-0.04 |
| 8.spy_vol_5d       | -0.21 +1.5   +0.07/+0.29   |
| 9.sc.held_5d       | +0.21 +0.009   +0.24/+0.03 |
| 10.top10_20d       | +0.20 +1.5   +0.10/-0.10   |
| 11.sc.top10_5d     | +0.18 +0.003   +0.03/-0.15 |
| 12.smh_20d         | +0.15 +0.7   -0.09/-0.24   |
| 13.sc.bottom10_20d | +0.15 +0.001   +0.15/-0.00 |
| 14.bottom10_20d    | +0.14 +0.3   -0.10/-0.24   |
| 15.spread_20d      | +0.14 +1.2   +0.15/+0.01   |
| 16.sc.top10_1d     | +0.14 +0.003   +0.02/-0.12 |
| 17.spy_20d         | +0.14 -0.0   -0.15/-0.29   |
| 18.book_1d         | -0.13 -0.2   -0.10/+0.03   |
| 19.qqq_vol_5d      | -0.13 +1.1   +0.13/+0.26   |
| 20.qqq_20d         | +0.13 +0.1   -0.13/-0.26   |
| 21.spy_vol_1d      | -0.12 +0.7   +0.06/+0.18   |
| 22.sc.bottom10_1d  | -0.12 -0.002   +0.10/+0.22 |
| 23.psq_20d         | -0.11 -0.1   +0.15/+0.26   |
| 24.spy_1d          | -0.11 -0.1   -0.07/+0.04   |
| 25.sc.universe_1d  | +0.11 +0.000   -0.28/-0.39 |
| 26.psq_1d          | +0.09 +0.1   +0.05/-0.04   |
| 27.qqq_1d          | -0.09 -0.1   -0.05/+0.03   |
| 28.spread_5d       | +0.08 +0.3   +0.06/-0.01   |
| 29.spy_5d          | -0.07 -0.3   -0.17/-0.10   |
| 30.top10_1d        | -0.07 -0.1   -0.04/+0.03   |
| 31.book_5d         | -0.06 -0.2   -0.19/-0.12   |
| 32.soxs_5d         | +0.05 +1.2   +0.12/+0.07   |
| 33.soxs_1d         | +0.05 +0.3   -0.02/-0.07   |
| 34.spread_1d       | -0.05 -0.1   +0.03/+0.08   |
| 35.spy_vol_20d     | +0.05 +3.9   +0.13/+0.08   |
| 36.smh_1d          | -0.05 -0.1   +0.01/+0.06   |
| 37.universe_1d     | -0.05 -0.1   -0.05/-0.00   |
| 38.psq_5d          | +0.04 +0.3   +0.16/+0.12   |
| 39.sc.bottom10_5d  | -0.04 +0.000   +0.11/+0.15 |
| 40.qqq_vol_1d      | -0.03 +0.7   +0.09/+0.13   |
| 41.qqq_5d          | -0.03 -0.3   -0.15/-0.12   |
| 42.bottom10_5d     | +0.03 -0.5   -0.14/-0.17   |
| 43.top10_5d        | -0.03 -0.3   -0.09/-0.06   |
| 44.universe_5d     | +0.02 -0.4   -0.14/-0.16   |
| 45.soxs_20d        | -0.02 -1.0   +0.16/+0.18   |
| 46.qqq_vol_20d     | +0.01 +2.8   +0.20/+0.19   |
| 47.bottom10_1d     | +0.01 +0.0   -0.08/-0.09   |
| 48.smh_5d          | -0.00 -0.2   -0.10/-0.10   |

FWD 20D (thr +/-7.5%)

| factor             | Dz (absGap   zUP/zDN)      |
|--------------------|----------------------------|
| 1.book_20d         | +0.54 +3.5   +0.25/-0.29   |
| 2.sc.top10_5d      | +0.46 +0.008   +0.11/-0.35 |
| 3.sc.top10_20d     | +0.42 +0.001   +0.10/-0.33 |
| 4.sc.held_5d       | +0.31 +0.005   +0.14/-0.17 |
| 5.sc.top10_1d      | +0.30 +0.007   +0.11/-0.18 |
| 6.sc.held_1d       | +0.25 +0.007   +0.14/-0.11 |
| 7.universe_20d     | +0.24 +1.2   +0.12/-0.13   |
| 8.spy_vol_20d      | +0.23 +2.8   +0.11/-0.12   |
| 9.sc.bottom10_20d  | +0.22 +0.002   +0.22/+0.00 |
| 10.top10_20d       | +0.21 +2.6   +0.26/+0.05   |
| 11.sc.held_20d     | +0.20 -0.007   +0.18/-0.02 |
| 12.bottom10_20d    | +0.18 +2.3   +0.03/-0.15   |
| 13.sc.universe_20d | -0.17 +0.000   -0.61/-0.44 |
| 14.smh_20d         | +0.17 +1.4   -0.04/-0.20   |
| 15.spy_vol_5d      | +0.14 +2.8   +0.00/-0.14   |
| 16.sc.universe_5d  | -0.14 +0.000   -0.46/-0.33 |
| 17.spread_5d       | +0.13 +0.7   +0.14/+0.01   |
| 18.book_5d         | +0.13 +0.5   +0.13/+0.00   |
| 19.psq_20d         | -0.10 -0.8   +0.04/+0.14   |
| 20.sc.bottom10_5d  | +0.10 +0.002   +0.13/+0.03 |
| 21.qqq_20d         | +0.09 +0.7   -0.03/-0.12   |
| 22.soxs_5d         | +0.08 +0.5   +0.02/-0.07   |
| 23.bottom10_5d     | -0.07 -0.4   -0.04/+0.03   |
| 24.sc.bottom10_1d  | -0.07 -0.001   +0.07/+0.14 |
| 25.spy_5d          | -0.07 -0.1   -0.01/+0.06   |
| 26.bottom10_1d     | +0.07 +0.1   +0.01/-0.06   |
| 27.spy_vol_1d      | +0.06 +2.2   +0.03/-0.03   |
| 28.spread_1d       | -0.05 -0.1   +0.07/+0.12   |
| 29.universe_1d     | +0.05 +0.0   +0.05/-0.00   |
| 30.qqq_5d          | -0.04 -0.1   -0.02/+0.02   |
| 31.psq_5d          | +0.04 +0.1   +0.02/-0.02   |
| 32.sc.universe_1d  | -0.04 +0.000   -0.36/-0.32 |
| 33.smh_5d          | -0.03 +0.0   +0.01/+0.05   |
| 34.qqq_vol_20d     | +0.03 +0.8   +0.16/+0.12   |
| 35.smh_1d          | +0.03 +0.1   +0.05/+0.02   |
| 36.soxs_20d        | -0.03 -2.9   +0.10/+0.13   |
| 37.qqq_vol_1d      | -0.03 +0.1   +0.01/+0.03   |
| 38.book_1d         | +0.03 +0.1   +0.09/+0.06   |
| 39.universe_5d     | -0.03 -0.3   +0.03/+0.05   |
| 40.spy_20d         | +0.03 +0.5   -0.02/-0.04   |
| 41.top10_5d        | +0.02 +0.2   +0.14/+0.12   |
| 42.soxs_1d         | -0.02 -0.1   -0.04/-0.02   |
| 43.spy_1d          | +0.02 +0.0   +0.02/+0.00   |
| 44.psq_1d          | -0.02 -0.0   -0.02/+0.00   |
| 45.qqq_1d          | +0.02 +0.0   +0.02/+0.00   |
| 46.top10_1d        | -0.01 -0.0   +0.09/+0.11   |
| 47.spread_20d      | -0.01 +0.4   +0.07/+0.08   |
| 48.qqq_vol_5d      | -0.00 +0.6   -0.01/-0.01   |

# Normalized trailing(z) -> raw forward correlation call-outs (|r| >= 0.5)

SINCE COVID (2020-04-01 to 2026-06-18, n=1562) | z = rolling z-score (value - recent mean)/recent std, L=63d

z=mean trailing z in panel; avg=mean raw forward return. Small-n panels (n<~40) noise-dominated.

## Forward 1-Day portfolio return (filtered by book\_fwd\_1d)

UP > +2.0% (n=230)

(none |r|>=0.5)

UP > +3.5% (n=52)

MODERATE (0.50-0.70):

|       |                |         |    |                 |            |
|-------|----------------|---------|----|-----------------|------------|
| +0.55 | universe_5d    | z=-0.26 | -> | spread_fwd_1d   | avg=+0.6%  |
| -0.53 | soxs_5d        | z=+0.12 | -> | spread_fwd_1d   | avg=+0.6%  |
| +0.52 | book_5d        | z=-0.10 | -> | spread_fwd_1d   | avg=+0.6%  |
| +0.52 | top10_5d       | z=-0.07 | -> | spread_fwd_1d   | avg=+0.6%  |
| +0.51 | book_1d        | z=-0.09 | -> | soxs_fwd_5d     | avg=-12.6% |
| +0.51 | sc.bottom10_5d | z=-0.26 | -> | bottom10_fwd_5d | avg=+2.6%  |
| +0.50 | spy_vol_20d    | z=+0.18 | -> | smh_fwd_1d      | avg=+3.6%  |
| +0.50 | qqq_5d         | z=-0.25 | -> | spread_fwd_1d   | avg=+0.6%  |
| -0.50 | psq_5d         | z=+0.27 | -> | spread_fwd_1d   | avg=+0.6%  |

DOWN < -2.0% (n=183)

(none |r|>=0.5)

DOWN < -3.5% (n=56)

MODERATE (0.50-0.70):

|       |              |         |    |               |           |
|-------|--------------|---------|----|---------------|-----------|
| -0.50 | universe_20d | z=-0.37 | -> | spread_fwd_1d | avg=-0.3% |
|-------|--------------|---------|----|---------------|-----------|

## Forward 5-Day portfolio return (filtered by book\_fwd\_5d)

UP > +4.5% (n=269)

(none |r|>=0.5)

UP > +6.5% (n=104)

MODERATE (0.50-0.70):

|       |             |         |    |                |           |
|-------|-------------|---------|----|----------------|-----------|
| +0.51 | sc.top10_5d | z=+0.10 | -> | spread_fwd_20d | avg=+1.8% |
|-------|-------------|---------|----|----------------|-----------|

DOWN < -4.5% (n=162)

(none |r|>=0.5)

DOWN < -6.5% (n=72)

MODERATE (0.50-0.70):

|       |            |         |    |                |           |
|-------|------------|---------|----|----------------|-----------|
| +0.58 | sc.held_5d | z=+0.13 | -> | spread_fwd_20d | avg=-2.9% |
|-------|------------|---------|----|----------------|-----------|

## Forward 20-Day portfolio return (filtered by book\_fwd\_20d)

UP > +7.5% (n=422)

(none |r|>=0.5)

UP > +12.5% (n=209)

(none |r|>=0.5)

DOWN < -7.5% (n=161)

MODERATE (0.50-0.70):

|       |              |         |    |                |           |
|-------|--------------|---------|----|----------------|-----------|
| +0.59 | spy_20d      | z=-0.04 | -> | spread_fwd_20d | avg=-2.5% |
| +0.54 | universe_20d | z=-0.13 | -> | spread_fwd_20d | avg=-2.5% |

DOWN < -12.5% (n=42)

STRONG (|r|>=0.70):

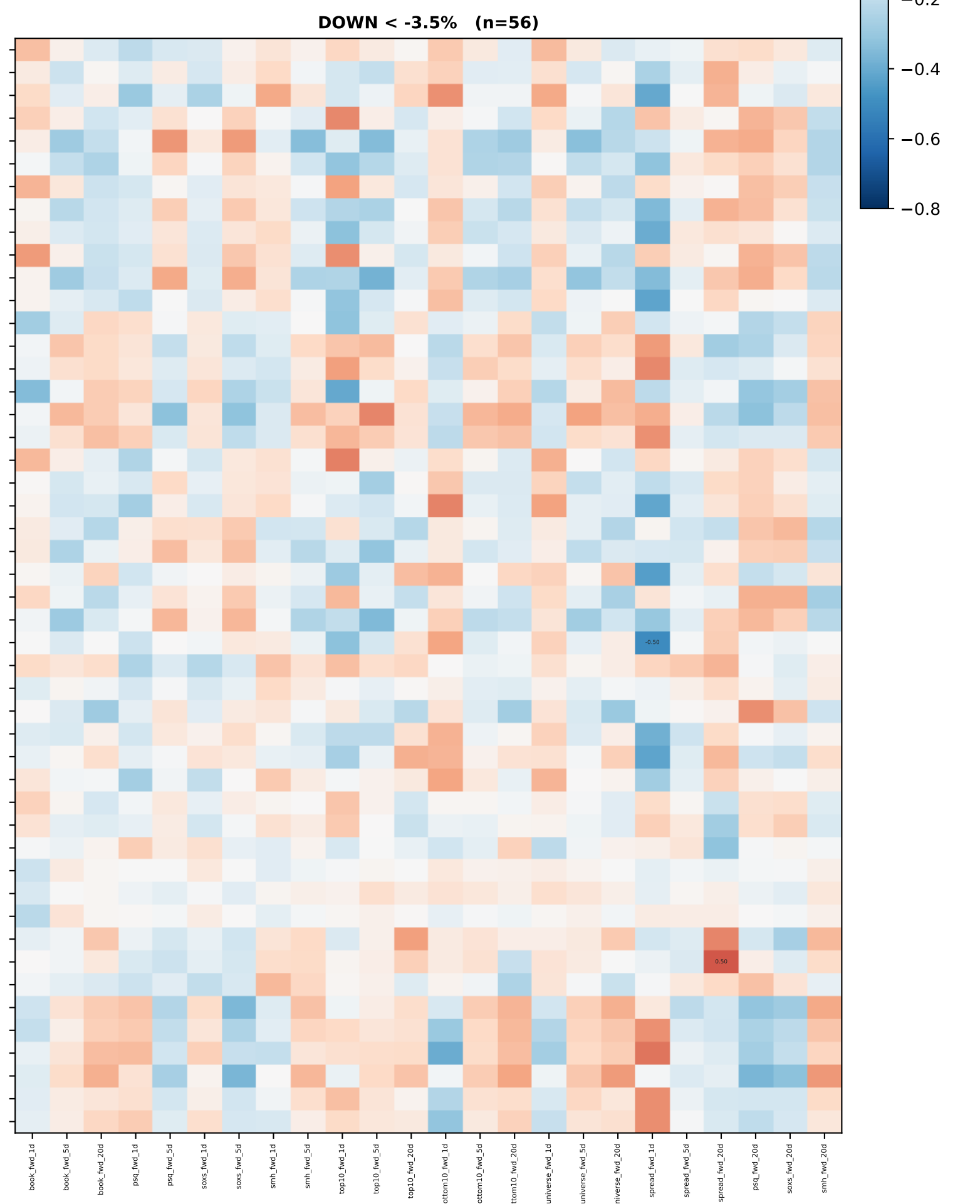
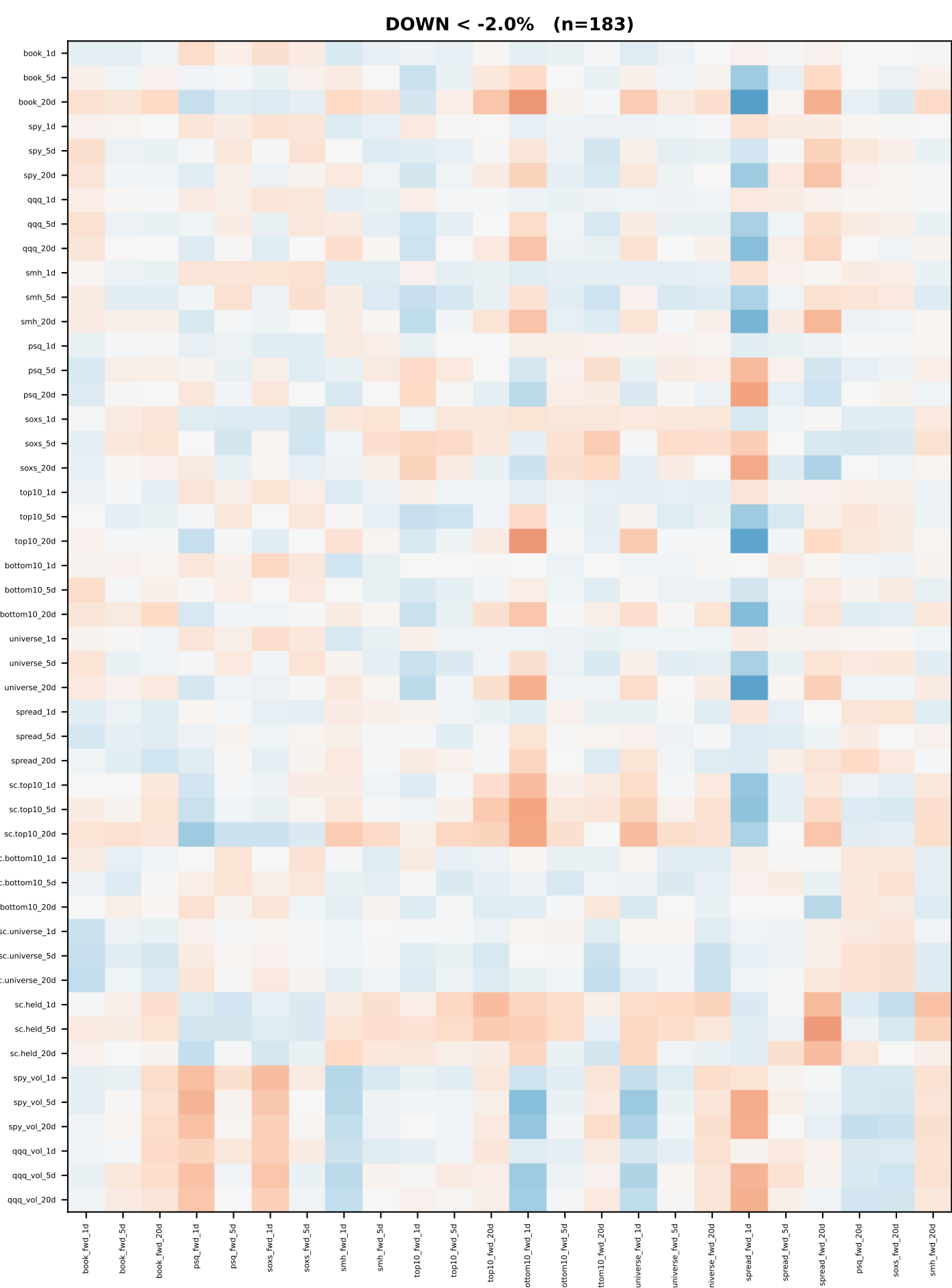
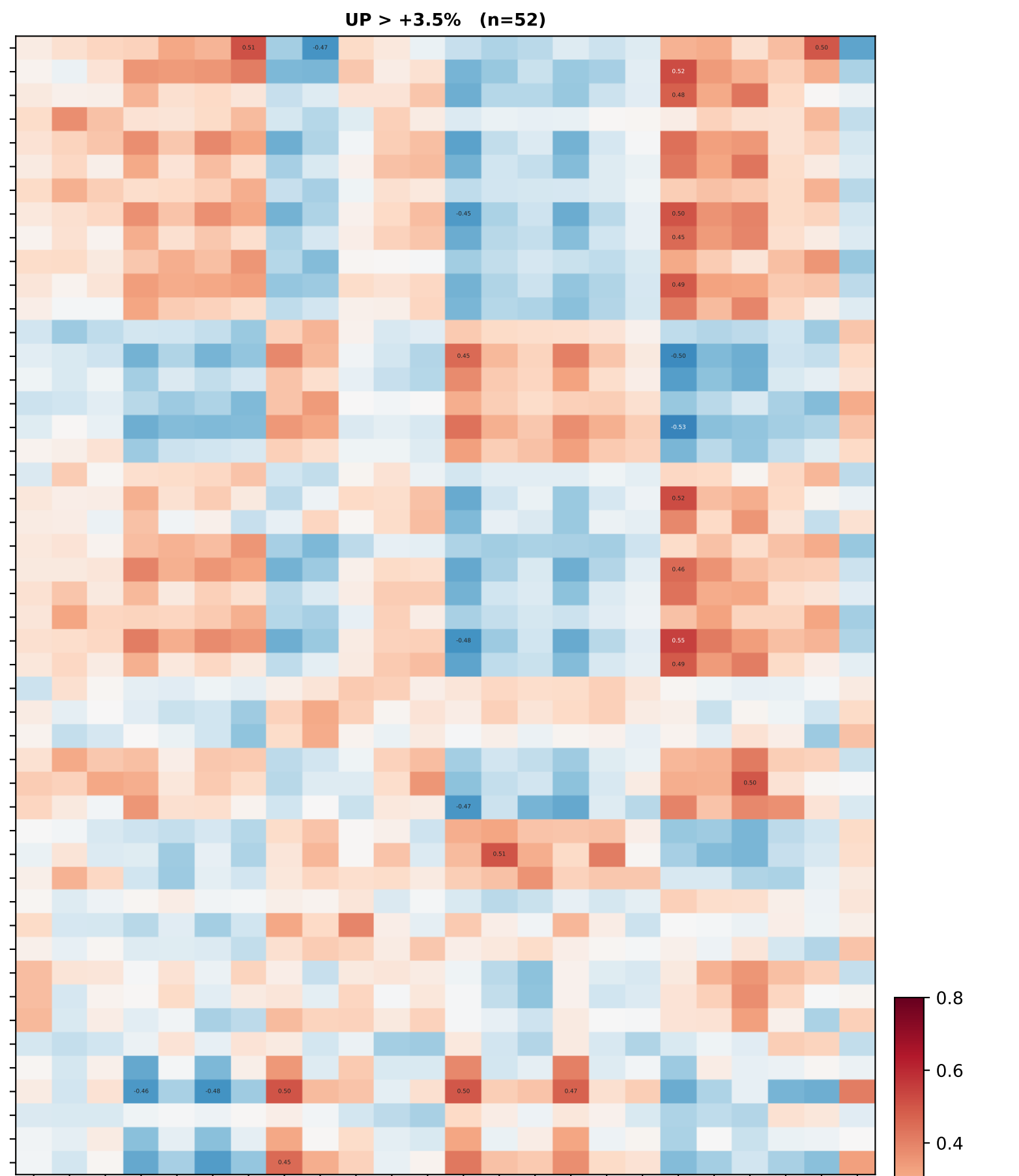
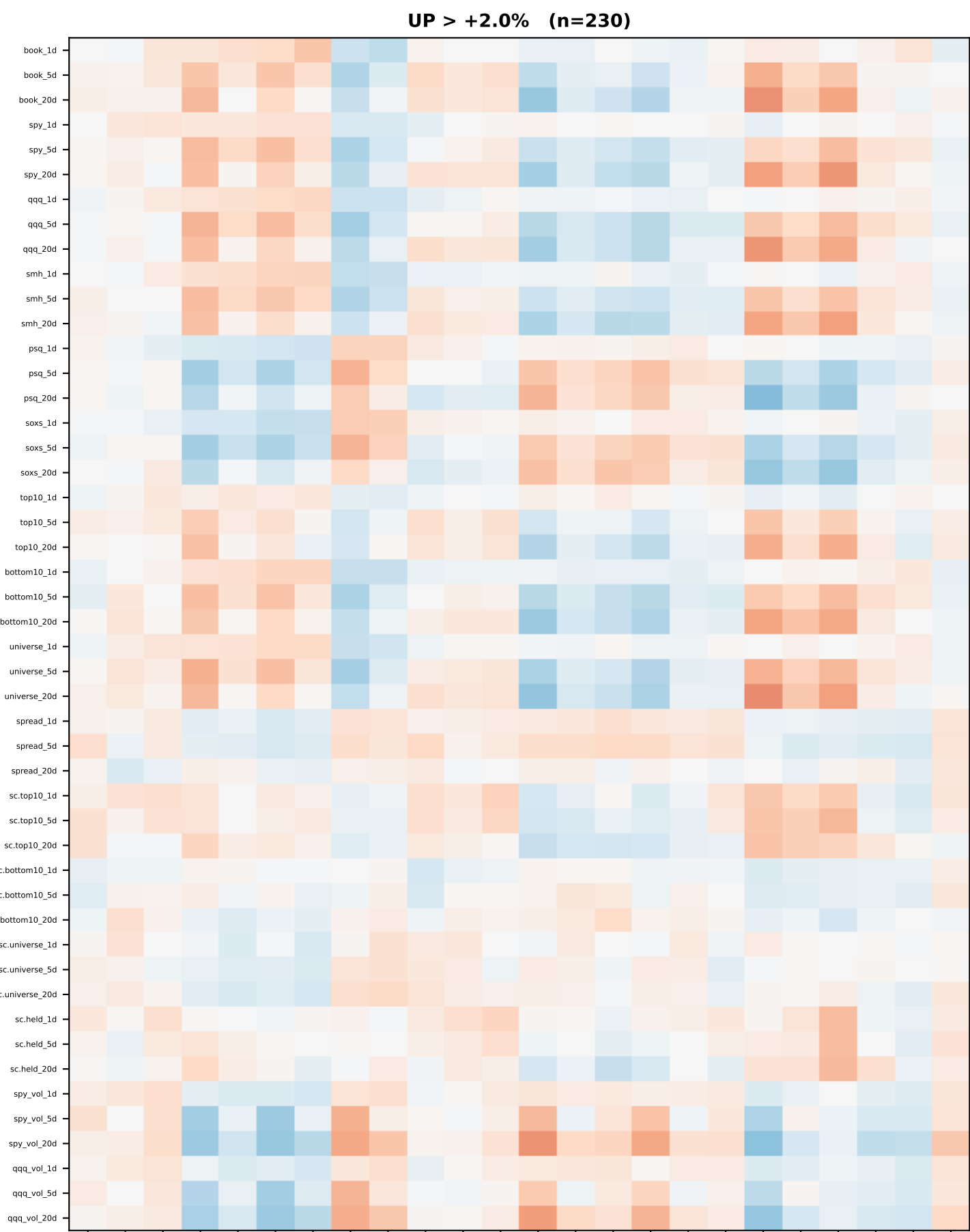
|       |         |         |    |                |           |
|-------|---------|---------|----|----------------|-----------|
| +0.79 | spy_20d | z=-0.33 | -> | spread_fwd_20d | avg=-2.4% |
| -0.71 | psq_20d | z=+0.45 | -> | spread_fwd_20d | avg=-2.4% |

MODERATE (0.50-0.70):

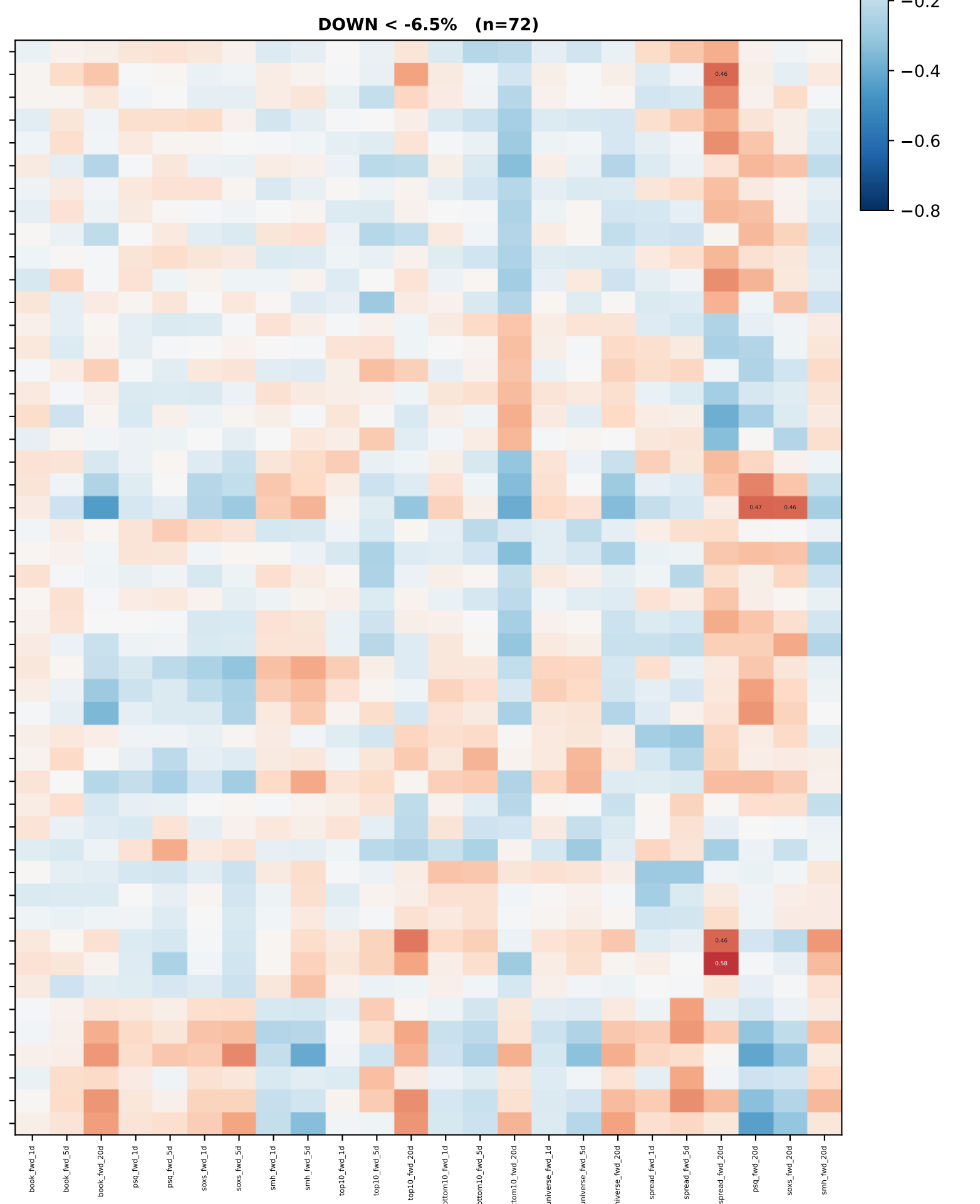
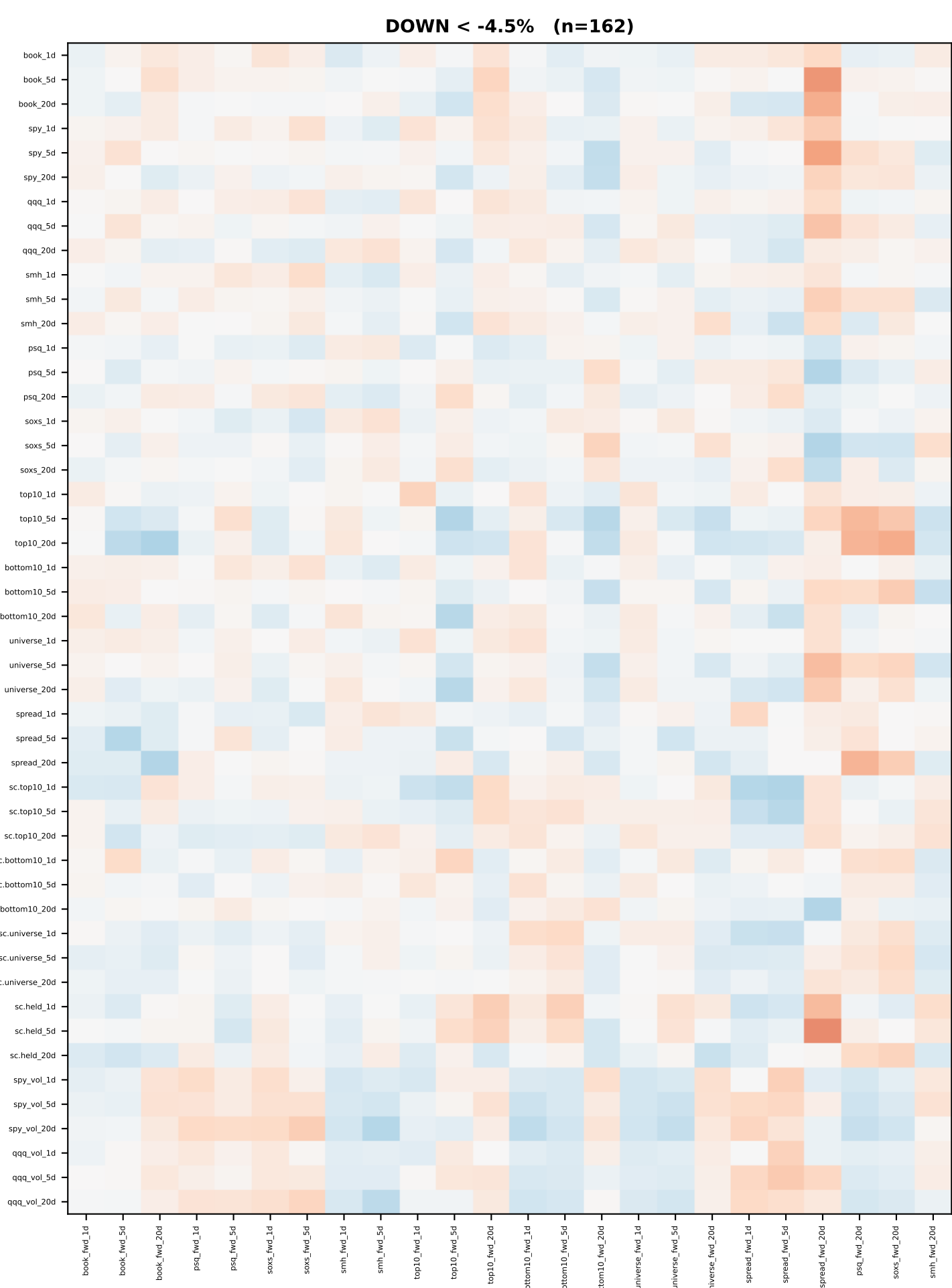
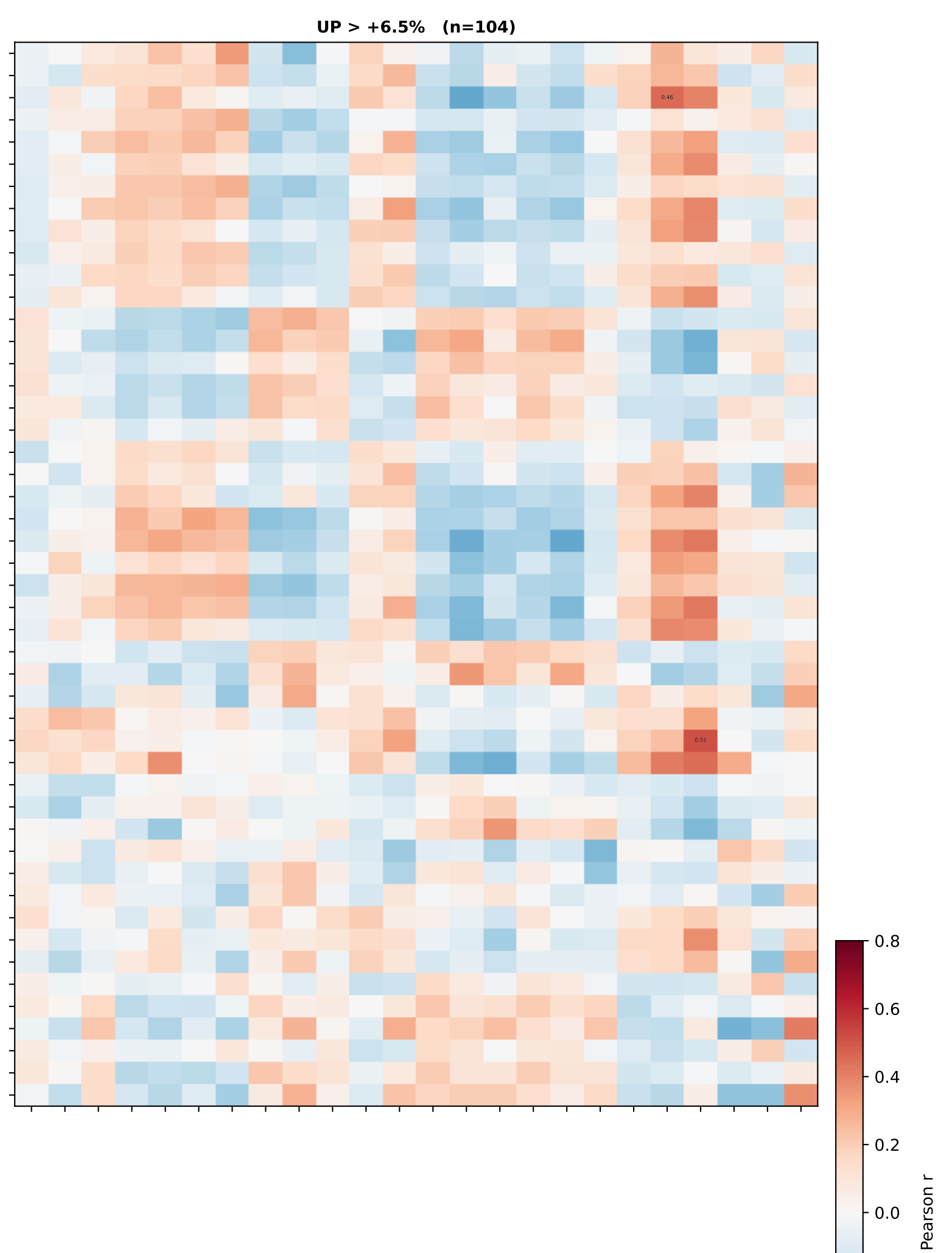
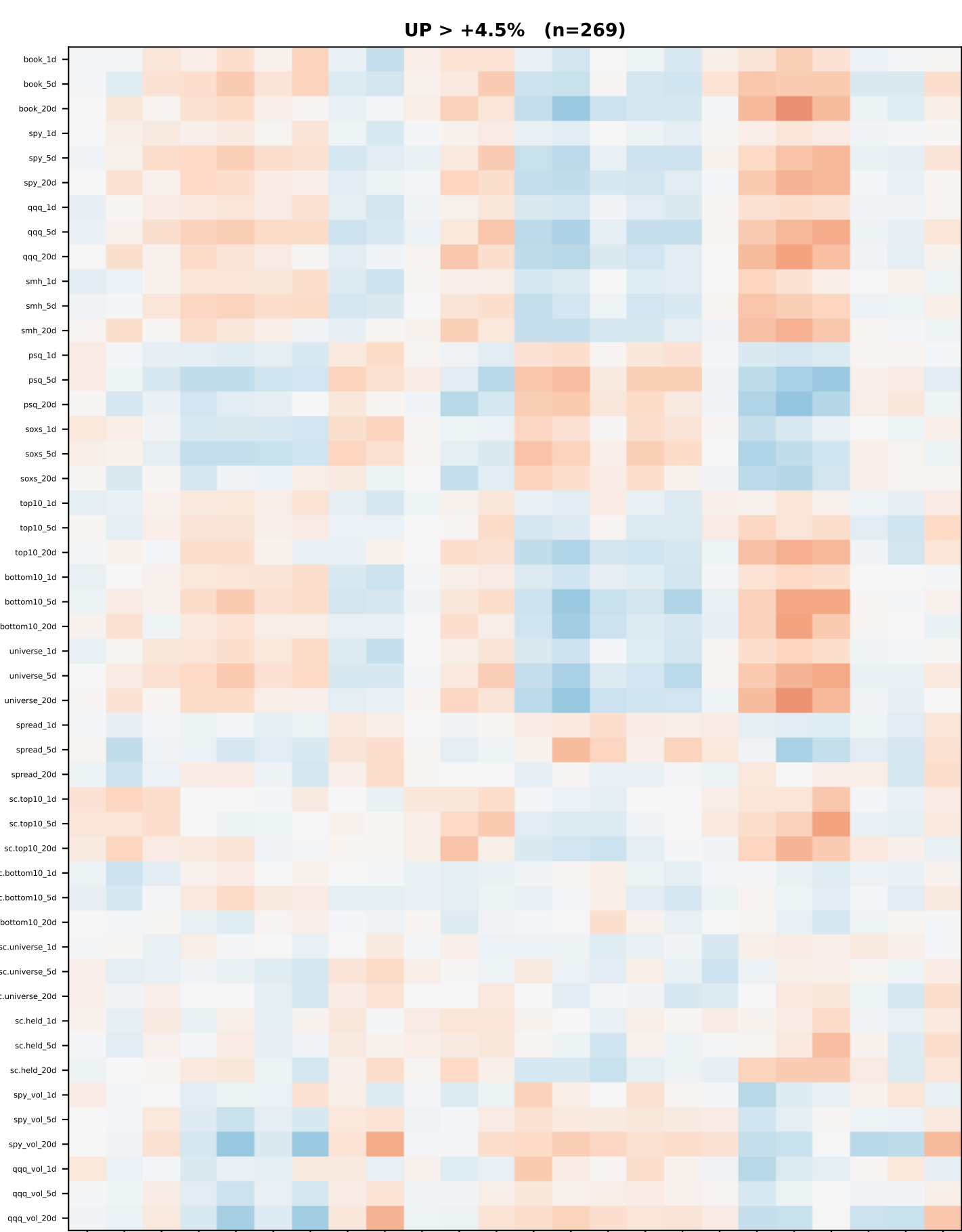
|       |                 |         |    |                  |            |
|-------|-----------------|---------|----|------------------|------------|
| +0.69 | qqq_20d         | z=-0.43 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.67 | spy_20d         | z=-0.33 | -> | bottom10_fwd_20d | avg=-12.2% |
| +0.66 | universe_20d    | z=-0.42 | -> | spread_fwd_20d   | avg=-2.4%  |
| +0.66 | psq_5d          | z=-0.05 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.66 | qqq_5d          | z=+0.05 | -> | bottom10_fwd_20d | avg=-12.2% |
| +0.66 | book_5d         | z=-0.12 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.66 | sc.bottom10_20d | z=-0.07 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.64 | sc.top10_20d    | z=-0.25 | -> | book_fwd_20d     | avg=-16.4% |

|       |                 |         |    |                  |            |
|-------|-----------------|---------|----|------------------|------------|
| +0.64 | bottom10_20d    | z=-0.47 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.64 | soxs_20d        | z=+0.53 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.63 | sc.bottom10_20d | z=-0.07 | -> | psq_fwd_20d      | avg=+9.9%  |
| +0.62 | spy_20d         | z=-0.33 | -> | spread_fwd_5d    | avg=+0.5%  |
| +0.62 | sc.bottom10_20d | z=-0.07 | -> | bottom10_fwd_20d | avg=-12.2% |
| +0.61 | book_20d        | z=-0.68 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.61 | spy_5d          | z=+0.05 | -> | bottom10_fwd_20d | avg=-12.2% |
| +0.61 | qqq_5d          | z=+0.05 | -> | spread_fwd_20d   | avg=-2.4%  |
| +0.61 | smh_20d         | z=-0.67 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.61 | psq_5d          | z=-0.05 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.61 | book_5d         | z=-0.12 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.60 | smh_5d          | z=+0.13 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.60 | spread_5d       | z=+0.21 | -> | book_fwd_20d     | avg=-16.4% |
| -0.59 | spy_vol_5d      | z=+0.12 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.59 | sc.top10_5d     | z=-0.50 | -> | book_fwd_20d     | avg=-16.4% |
| +0.58 | soxs_5d         | z=-0.11 | -> | universe_fwd_20d | avg=-12.2% |
| +0.57 | spy_5d          | z=+0.05 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.57 | smh_5d          | z=+0.13 | -> | universe_fwd_20d | avg=-12.2% |
| +0.57 | psq_20d         | z=+0.45 | -> | bottom10_fwd_20d | avg=-12.2% |
| +0.57 | soxs_5d         | z=-0.11 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.56 | qqq_20d         | z=-0.43 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.56 | universe_5d     | z=-0.03 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.56 | qqq_vol_5d      | z=+0.02 | -> | spread_fwd_20d   | avg=-2.4%  |
| +0.56 | psq_5d          | z=-0.05 | -> | universe_fwd_20d | avg=-12.2% |
| +0.55 | bottom10_20d    | z=-0.47 | -> | spread_fwd_5d    | avg=+0.5%  |
| -0.55 | qqq_5d          | z=+0.05 | -> | universe_fwd_20d | avg=-12.2% |
| +0.55 | sc.bottom10_5d  | z=+0.29 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.55 | top10_5d        | z=+0.27 | -> | book_fwd_20d     | avg=-16.4% |
| +0.54 | universe_5d     | z=-0.03 | -> | spread_fwd_20d   | avg=-2.4%  |
| -0.54 | sc.bottom10_20d | z=-0.07 | -> | soxs_fwd_20d     | avg=+35.2% |
| +0.54 | sc.bottom10_20d | z=-0.07 | -> | universe_fwd_20d | avg=-12.2% |
| +0.54 | soxs_20d        | z=+0.53 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.54 | smh_20d         | z=-0.67 | -> | bottom10_fwd_20d | avg=-12.2% |
| +0.53 | smh_5d          | z=+0.13 | -> | spread_fwd_20d   | avg=-2.4%  |
| +0.53 | sc.held_20d     | z=+0.02 | -> | smh_fwd_20d      | avg=-12.6% |
| +0.53 | spy_20d         | z=-0.33 | -> | psq_fwd_20d      | avg=+9.9%  |
| +0.53 | book_20d        | z=-0.68 | -> | spread_fwd_5d    | avg=+0.5%  |
| -0.53 | bottom10_20d    | z=-0.47 | -> | book_fwd_20d     | avg=-16.4% |
| -0.53 | sc.bottom10_5d  | z=+0.29 | -> | spread_fwd_20d   | avg=-2.4%  |
| +0.52 | bottom10_20d    | z=-0.47 | -> | top10_fwd_20d    | avg=-14.6% |
| -0.52 | psq_20d         | z=+0.45 | -> | spread_fwd_5d    | avg=+0.5%  |
| +0.52 | soxs_5d         | z=-0.11 | -> | smh_fwd_20d      | avg=-12.6% |
| -0.52 | sc.bottom10_5d  | z=+0.29 | -> | psq_fwd_20d      | avg=+9.9%  |
| -0.52 | universe_20d    | z=-0.42 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.51 | spy_20d         | z=-0.33 | -> | universe_fwd_20d | avg=-12.2% |
| +0.51 | universe_20d    | z=-0.42 | -> | spread_fwd_5d    | avg=+0.5%  |
| -0.51 | spy_5d          | z=+0.05 | -> | universe_fwd_20d | avg=-12.2% |
| +0.51 | sc.bottom10_5d  | z=+0.29 | -> | universe_fwd_20d | avg=-12.2% |
| +0.50 | qqq_20d         | z=-0.43 | -> | spread_fwd_5d    | avg=+0.5%  |
| -0.50 | top10_5d        | z=+0.27 | -> | bottom10_fwd_20d | avg=-12.2% |
| -0.50 | top10_5d        | z=+0.27 | -> | universe_fwd_20d | avg=-12.2% |
| +0.50 | book_20d        | z=-0.68 | -> | top10_fwd_20d    | avg=-14.6% |

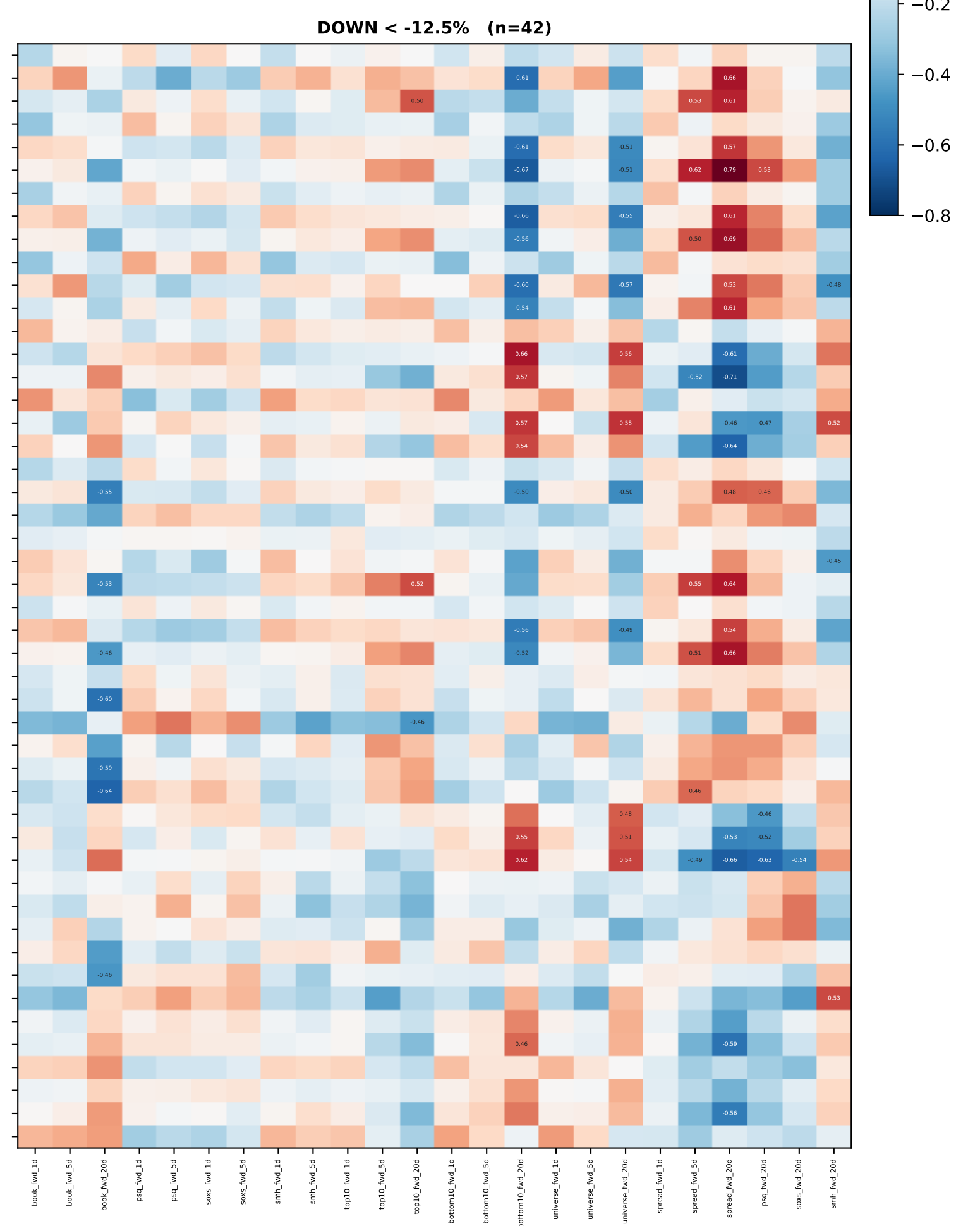
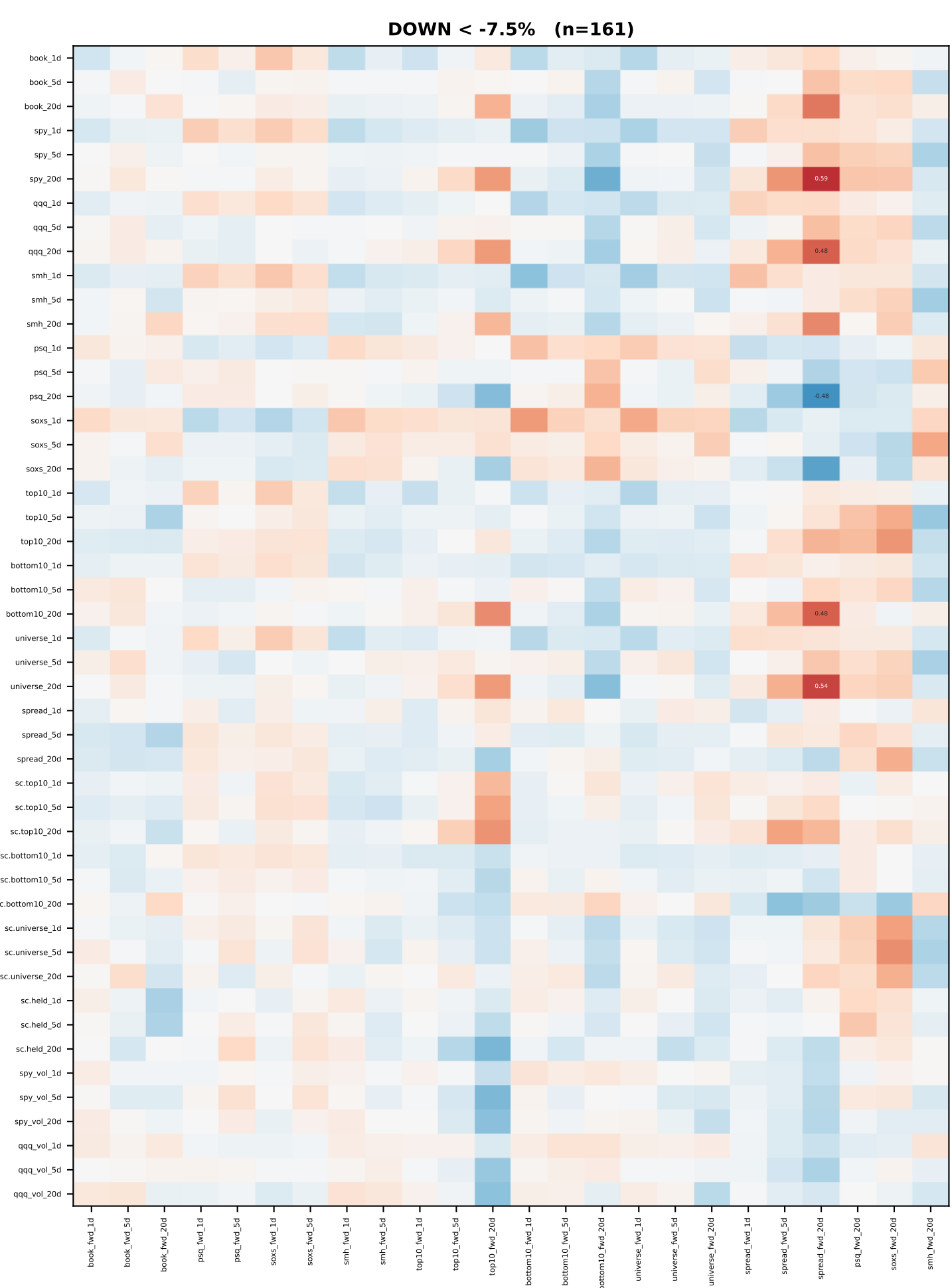
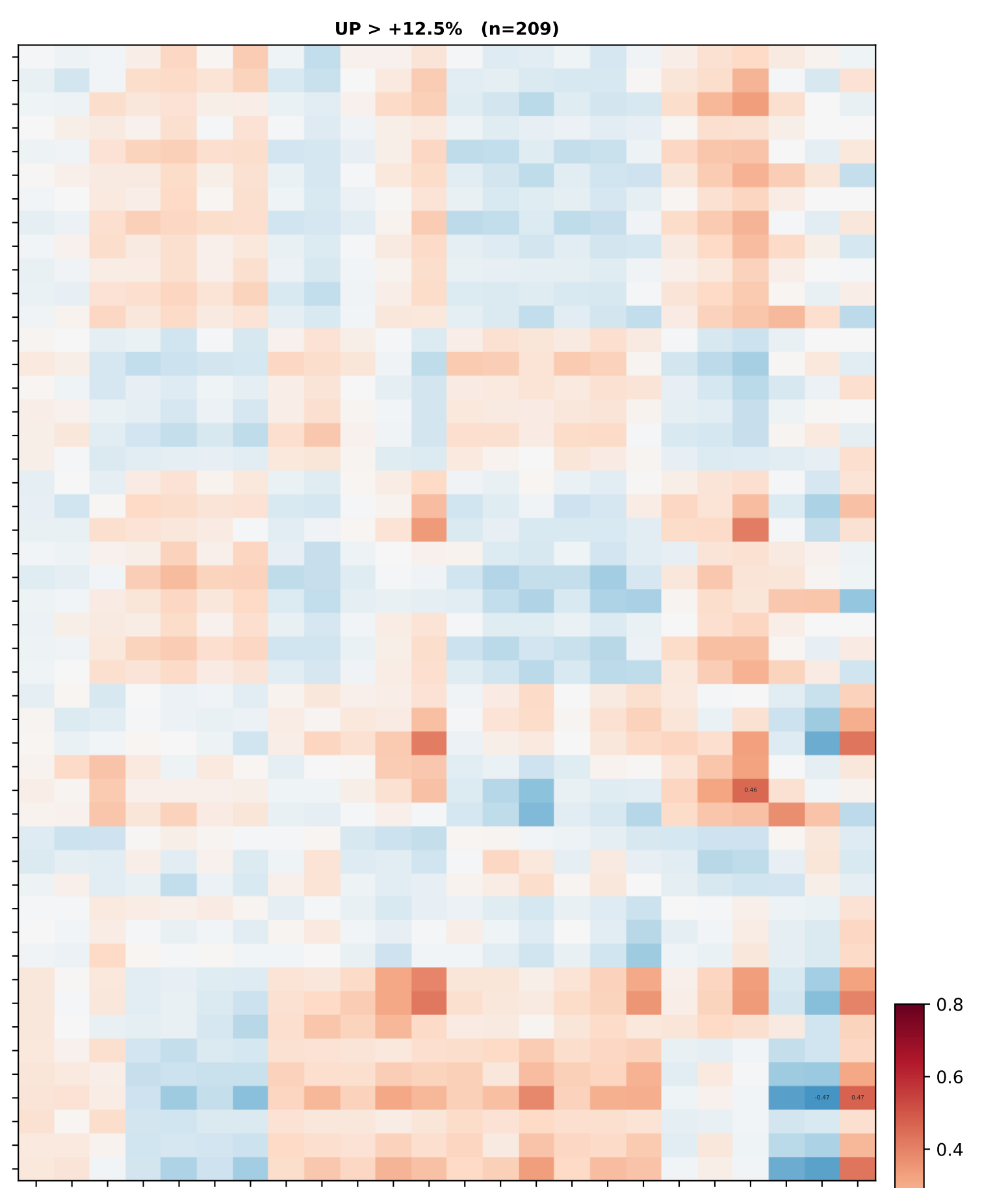
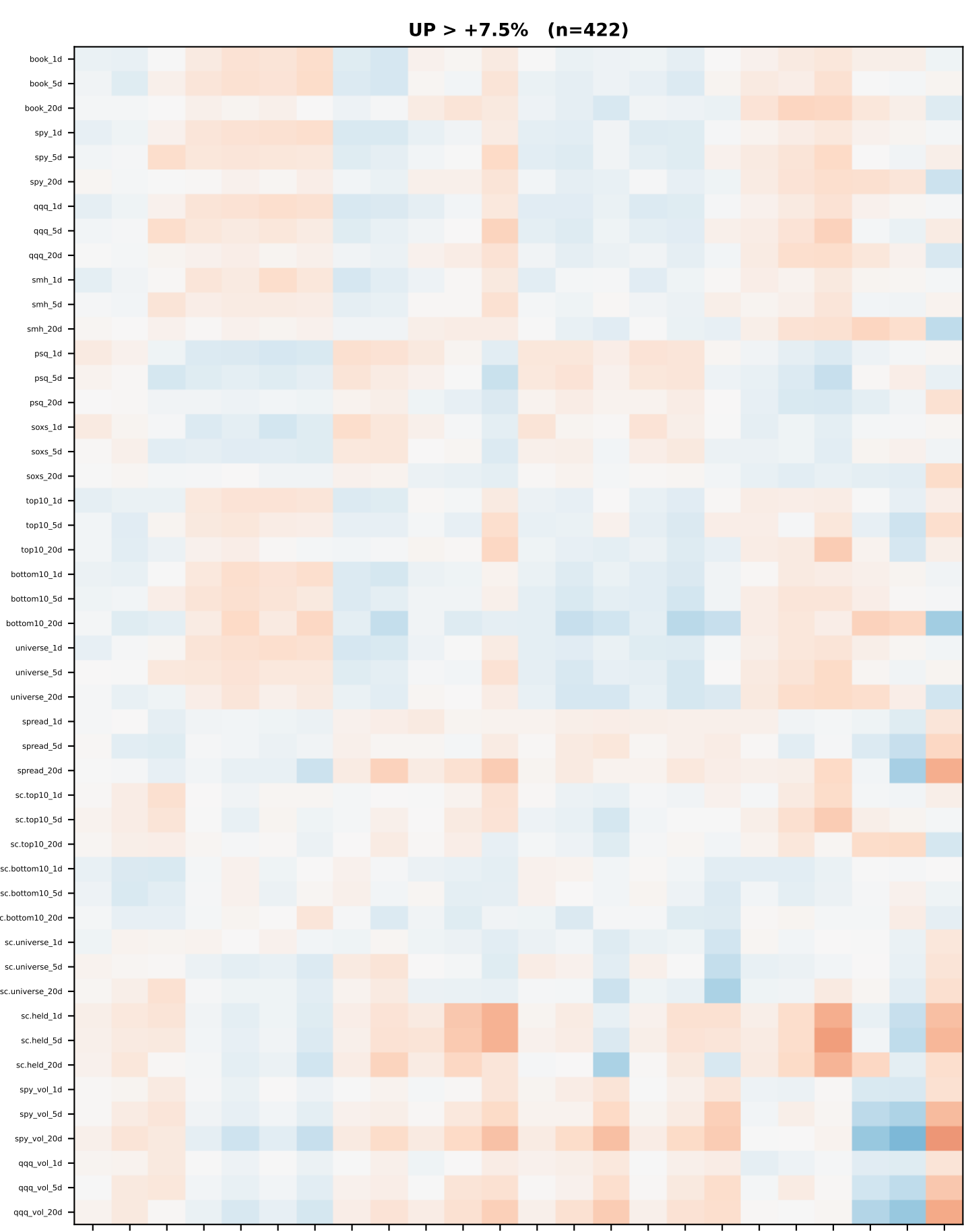




Pearson r







Pearson r